Web Performance Audit

NovaPay SpA — Audit Report

Version 1.0
March 2026
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Confidential — Vantx

WEB PERFORMANCE AUDIT

NovaPay SpA — app.novapay.com

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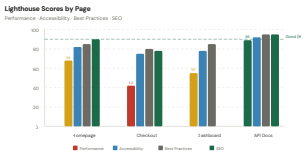
Executive Summary

NovaPay's web application (app.novapay.com) was audited across 4 key pages. The checkout page — the most critical revenue path — scored 42/100 on Lighthouse with a 4.2s LCP, primarily caused by an unoptimized hero image (2.4 MB PNG) and 1.2 MB of render-blocking JavaScript. The dashboard loads 3 analytics libraries synchronously, adding 890ms to first paint. Quick wins on P0 items alone would improve checkout from 42 to ~72 Lighthouse points and bring LCP under 2.5s.

Overall Scores

PAGE	PERFORMANCE	ACCESSIBILITY	BEST PRACTICES	SEO	STATUS
Homepage	68/100	82/100	85/100	90/100	⚠️ Needs work
Checkout	42/100	75/100	80/100	78/100	❌ Poor
Dashboard	55/100	78/100	85/100	N/A	⚠️ Needs work
API Docs	89/100	92/100	95/100	95/100	✅ Good



Lighthouse Scores by Page

1. Core Web Vitals

Lab Data (Lighthouse)

PAGE	LCP	INP	CLS	FCP	TTFB	SI
Homepage	2.1s	85ms	0.05	1.2s	180ms	2.4s
Checkout	4.2s	120ms	0.12	1.8s	190ms	4.8s
Dashboard	3.1s	95ms	0.08	1.5s	210ms	3.5s
API Docs	1.4s	45ms	0.02	0.8s	120ms	1.5s

Field Data (CrUX / RUM — Faro)

PAGE	LCP (P75)	INP (P75)	CLS (P75)	STATUS
Homepage	2.4s	92ms	0.06	⚠️ Needs improvement
Checkout	4.8s	145ms	0.15	❌ Poor
Dashboard	3.4s	105ms	0.09	⚠️ Needs improvement

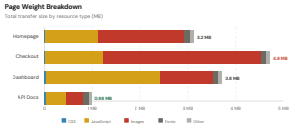
Targets

METRIC	GOOD	NEEDS IMPROVEMENT	POOR
LCP	≤2.5s	2.5–4.0s	>4.0s
INP	≤200ms	200–500ms	>500ms
CLS	≤0.1	0.1–0.25	>0.25

2. Resource Analysis

Page Weight

PAGE	TOTAL	HTML	CSS	JS	IMAGES	FONTS	OTHER
Homepage	3.2 MB	18 KB	95 KB	1.1 MB	1.8 MB	120 KB	62 KB
Checkout	4.8 MB	22 KB	110 KB	1.2 MB	3.3 MB	120 KB	48 KB
Dashboard	3.8 MB	15 KB	88 KB	2.4 MB	1.1 MB	120 KB	72 KB
API Docs	980 KB	45 KB	32 KB	420 KB	360 KB	120 KB	3 KB



Page Weight Breakdown

JavaScript Analysis

BUNDLE	SIZE	PARSED	COVERAGE	NOTES
`main.js`	680 KB	240ms	38% used	62% dead code — tree-shaking not configured
`vendor.js`	420 KB	180ms	52% used	Includes full lodash, moment.js (both replaceable)
`analytics.js` **(Segment)**	210 KB	95ms	100%	Render-blocking — should be deferred
`hotjar.js`	180 KB	80ms	100%	Render-blocking — should be async
`intercom.js`	310 KB	120ms	100%	Loaded on all pages — only needed on dashboard

Image Analysis

IMAGE	FORMAT	SIZE	OPTIMAL FORMAT	SAVINGS
checkout-hero.png	PNG	2.4 MB	WebP	-85% (→ 360 KB)
homepage-hero.jpg	JPEG	1.2 MB	WebP + responsive	-70% (→ 360 KB)
trust-badges.png	PNG	450 KB	SVG	-95% (→ 22 KB)
merchant-logos.png	PNG sprite	680 KB	Individual SVGs	-90% (→ 68 KB)

Third-Party Impact

SCRIPT	SIZE	BLOCK TIME	PURPOSE	RECOMMENDATION
Segment Analytics	210 KB	95ms	Analytics	Defer load, use `analytics.load()` after FCP
Hotjar	180 KB	80ms	Heatmaps	Async, load only on marketing pages
Intercom	310 KB	120ms	Chat widget	Lazy-load on dashboard only, remove from checkout
Stripe.js	45 KB	15ms	Payments	Keep — required for checkout, already optimized

3. Loading Sequence

Critical Rendering Path

RESOURCE	TYPE	SIZE	PRIORITY	TIMING	ISSUE
`index.html`	Document	22 KB	Highest	0–190ms	OK
`main.css`	Stylesheet	110 KB	Highest	190–310ms	OK — critical CSS needed
`vendor.js`	Script	420 KB	High	190–480ms	**Render-blocking**
`main.js`	Script	680 KB	High	190–620ms	**Render-blocking**
`analytics.js`	Script	210 KB	High	190–520ms	**Render-blocking — should be deferred**
`checkout-hero.png`	Image	2.4 MB	Low	620–2800ms	**LCP element — delayed by JS**
Fonts (DM Sans)	Font	120 KB	High	310–450ms	OK — preloaded

Render-Blocking Resources

RESOURCE	TYPE	SIZE	FIX
`vendor.js`	Script	420 KB	Add `defer` attribute, move to bottom of `<body>`
`main.js`	Script	680 KB	Add `defer`, implement code-splitting (route-based)
`analytics.js`	Script	210 KB	Add `defer` + load asynchronously after FCP

RESOURCE	TYPE	SIZE	FIX
`hotjar.js`	Script	180 KB	Add `async`, conditionally load

4. Findings & Recommendations

ID	SEVERITY	CATEGORY	FINDING	FIX	IMPACT
WP-01	P0	Images	Checkout hero is 2.4 MB PNG	Convert to WebP, add srcset for responsive sizes	LCP -1.5s, page weight -2 MB
WP-02	P0	JavaScript	1.2 MB of render-blocking JS on checkout	Add `defer` to all non-critical scripts	FCP -400ms, LCP -800ms
WP-03	P0	Third-party	Analytics + Hotjar block rendering on all pages	Defer analytics, async Hotjar, conditionally load Intercom	FCP -300ms
WP-04	P1	JavaScript	62% dead code in main.js (tree-shaking disabled)	Enable tree-shaking, replace lodash with lodash-es, drop moment.js for date-fns	JS size -350 KB
WP-05	P1	Images	All images served as PNG/JPEG — no modern formats	Implement `<picture>` with WebP/AVIF + fallback	Total image savings -75%
WP-06	P1	CLS	Checkout images have no explicit dimensions	Add width/height to all `<img>` tags, use aspect-ratio CSS	CLS from 0.12 to <0.05
WP-07	P2	Caching	Static assets served with 1-hour cache	Set `Cache-Control: public, max-age=31536000,`	Repeat visits -60% faster

ID	SEVERITY	CATEGORY	FINDING	FIX	IMPACT
				`immutable` for hashed assets	
WP-08	P2	Fonts	Font loaded from Google Fonts (external)	Self-host fonts, add `font-display: swap`, preload critical variant	FOUT eliminated, -50ms TTFB

5. Remediation Roadmap

P0 — Immediate (estimated: +30 Lighthouse points)

WP-01: Convert checkout-hero.png to WebP. Generate responsive sizes (480w, 768w, 1200w). Use `<picture>` element with WebP + JPEG fallback. Expected: LCP from 4.2s to ~2.7s.

WP-02: Add `defer` attribute to `vendor.js` and `main.js`. Move `<script>` tags to end of `<body>`. Implement route-based code splitting for `/checkout` vs `/dashboard`. Expected: FCP from 1.8s to ~1.1s.

WP-03: Replace synchronous Segment load with `analytics.load()` called after `DOMContentLoaded`. Add `async` to Hotjar. Remove Intercom from checkout page entirely. Expected: -300ms FCP across all pages.

P1 — This sprint

WP-04: Enable tree-shaking in Webpack config. Replace `lodash` with `lodash-es` (enables per-function imports). Replace `moment.js` with `date-fns`. Run `webpack-bundle-analyzer` to find remaining dead code.

WP-05: Set up image pipeline: source → WebP + AVIF + original. Use `<picture>` for all above-fold images. Lazy-load below-fold images with `loading="lazy"`.

WP-06: Audit all `<img>` tags. Add explicit `width` and `height`. Add `aspect-ratio` CSS for responsive images.

P2 — This quarter

WP-07: Update CDN config for immutable asset caching. **WP-08:** Self-host fonts with `font-display: swap`.

Projected Improvement

PAGE	CURRENT SCORE	AFTER P0	AFTER P1	TARGET
Homepage	68/100	82/100	90/100	90+
Checkout	42/100	72/100	85/100	85+
Dashboard	55/100	70/100	82/100	80+
API Docs	89/100	89/100	92/100	90+

Next steps: Implement P0 fixes → VANTX re-audits for free (included) → Validate improvement with before/after Lighthouse comparison.

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